

SOOHYUN LEE

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Education

Korea Aerospace University Expected February 2027
Bachelor of Software Engineering GPA: 4.14/4.5

Research Interests

Trustworthy AI for robot behavior generation and verification

Publications

- **[LLM-Based Behavior Tree Generation for Long-Horizon Robot Task Planning]**. Accepted at *ICROS 2026*.
- **[LLM-Based Behavior Tree Generation for Mission Planning: A case Study on Aircraft Approach and Landing]**. Submitted to *KSAS 2026*.
- **[The Impact of Tree Structure and Construction Strategy on LLM-Based Behavior Tree Planning]**. Submitted to ICRA 2027.

Research Experience

Korea Aerospace University Computer Vision Lab June 2025 – September 2025
Undergraduate Research Student, Advisor: Sunok Kim

- Participated in weekly paper reading seminars and presented recent research in computer vision.

Korea Aerospace University Natural Language Processing Lab December 2025 – Present
Undergraduate Research Student, Advisor: Hyunyoung Kil

- Developed an encoder-decoder generative framework for cross-lingual ABSA, integrating instruction tuning, contrastive learning, reasoning, and multilingual/code-switch strategies.

Research Project in Collaboration with ROBOTIS March 2026 – Present
Undergraduate Research Project

- Evaluated forward/backward Behavior Tree structures and one-step/recursive LLM construction on 30 AI2-THOR household tasks across three task horizons, measuring task success and token usage with and without execution disturbances.
- In the reported aggregate results, one-step forward planning achieved 96.7% success without disturbances but 31.7% with disturbances; hybrid backward expansion improved success over one-step backward planning while using fewer tokens than recursive backward expansion.

Formal Verification of LLM-Based Behavior Trees April 2026 – Present
Undergraduate Researcher / RA-L Manuscript in Preparation

- Developed a formal proof of finite-time convergence for behavior trees in parallel multi-robot environments.

Saint Mary's University July 2026 – August 2026
Research Intern, Advisor: Iqbal Shameer

- Research on detecting abnormal heart murmurs in paediatric cardiology.

Awards & Scholarships

- Academic Excellence Scholarship, Korea Aerospace University (4/5 semesters)
- 2nd Place, University-Industry Collaboration Project, Korea Aerospace University July 2025
- 2nd Place, Adventure Design Course Project, Korea Aerospace University July 2024

Teaching & Activities

Teaching Assistant, Data Structures March 2025 – June 2025
Korea Aerospace University

- Graded programming assignments and provided feedback to support student learning.
- Assisted the instructor during lectures.

Teaching Assistant, Introduction to Coding September 2025 – December 2025
Korea Aerospace University

- Guided students through hands-on programming assignments.

Teaching Assistant, Computer Programming March 2026 – June 2026
Korea Aerospace University

- Supported students in completing foundational programming exercises.

Selected Coursework

Data Structures, Algorithms, Machine Learning, Deep Learning, Probability and Statistics

Skills

- **English:** TOEFL iBT 100
- **Programming:** Python, C++, Kotlin, Swift, C
- **AI/ML:** PyTorch, Pandas
- **Robotics:** ROS/ROS2, Behavior Trees
- **Tools:** Git, Linux, LaTeX, Docker